

FRANKFURT AM MAIN, Germany

WMO: 106370

Lat: 50.0258N

Lon: 8.5214E

Elev: 342

StdP: 14.52

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	21.4	8.2	8.5	20.5	13.6	11.1	24.6	27.8	49.3	24.6	47.7	7.0	40	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	89.7	67.9	86.0	66.6	82.4	65.1	70.1	83.4	68.6	80.5	67.3	78.2	7.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	98.5	72.3	64.7	92.8	71.5	63.4	88.3	71.1	34.3	83.5	33.1	80.3	32.0	78.5	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	18.8	16.6	12.0	95.9	6.1	2.7	7.6	97.8	4.0	99.4	0.6	100.9	-3.8	102.8		
(5)			10.9	73.0	5.9	1.3	6.7	74.0	3.2	74.7	-0.1	75.4	-4.4	76.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	35.6	37.5	44.1	52.2	58.5	65.5	69.4	67.6	60.4	52.1	43.3	37.7	(6)
(7)		DBStd	13.61	8.04	7.36	7.11	7.22	6.80	6.53	6.31	5.90	5.65	6.50	6.64	7.55	(7)
(8)		HDD50	1730	449	352	206	60	8	0	0	1	54	216	384		(8)
(9)		HDD65	5137	912	770	648	386	223	75	27	38	160	401	651	846	(9)
(10)		CDD50	2483	1	1	23	126	271	464	601	547	313	118	15	3	(10)
(11)		CDD65	419	0	0	0	2	21	89	163	120	23	1	0	0	(11)
(12)		CDH74	3883	0	0	0	58	238	814	1529	1051	185	8	0	0	(12)
(13)		CDH80	1310	0	0	0	4	51	257	593	363	42	0	0	0	(13)
(14)	Wind	WSAvg	7.7	8.6	8.4	8.8	7.9	7.7	7.5	7.4	6.8	7.0	7.1	7.3	8.0	(14)
(15)	Precipitation	PrecAvg	27.2	2.1	2.0	1.8	1.7	2.4	2.5	2.8	2.4	2.2	2.3	2.3	2.5	(15)
(16)		PrecMax	33.5	5.4	5.1	4.8	3.5	4.8	5.3	6.4	4.8	5.3	6.7	3.9	7.0	(16)
(17)		PrecMin	19.6	0.2	0.3	0.3	0.0	0.6	0.7	1.2	0.1	0.6	0.3	0.0	0.5	(17)
(18)		PrecStd	3.8	1.0	1.2	1.1	0.8	1.1	1.1	1.3	1.3	1.1	1.3	1.0	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	57.5	68.3	79.2	85.2	91.4	95.3	94.6	85.7	74.8	62.3	55.6	(19)
MCWB			51.3	48.7	53.2	58.5	66.2	68.9	69.5	69.3	66.1	62.2	56.2	50.9	(20)	
2%		DB	52.8	53.7	63.5	75.1	80.3	86.3	91.2	88.0	79.1	68.1	57.6	53.5	(21)	
		MCWB	49.0	47.4	51.0	57.1	63.5	67.0	68.3	66.5	64.4	60.1	53.3	49.7	(22)	
5%		DB	49.6	50.7	59.1	70.0	75.6	82.5	86.3	84.0	74.9	64.6	55.4	50.7	(23)	
		MCWB	46.8	46.1	49.6	55.6	60.5	65.5	66.5	65.6	63.0	58.5	51.7	47.7	(24)	
10%		DB	46.5	48.2	55.4	66.0	71.8	78.7	82.4	80.2	70.9	62.3	52.9	48.2	(25)	
		MCWB	44.2	44.3	48.3	53.7	58.3	64.1	65.1	64.8	61.0	57.2	50.0	45.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	51.3	55.5	60.9	68.5	72.4	71.7	71.5	68.7	63.6	57.6	52.8	(27)
MCD B			55.1	55.2	63.2	72.9	82.3	85.9	88.7	84.2	78.5	69.9	60.3	54.9	(28)	
2%		WB	49.5	48.6	53.1	58.3	65.3	69.3	70.0	69.6	66.4	61.4	54.6	50.4	(29)	
		MCD B	51.7	52.2	60.2	71.3	76.8	81.4	84.1	82.4	76.0	66.6	57.4	52.8	(30)	
5%		WB	46.6	46.7	51.1	56.3	62.5	67.4	68.7	67.8	64.3	59.6	52.2	48.1	(31)	
		MCD B	49.0	50.1	57.1	67.4	72.5	78.6	80.7	79.1	72.4	63.9	54.8	50.7	(32)	
10%		WB	44.1	44.4	48.9	54.7	60.1	65.7	67.3	66.3	62.5	57.8	50.2	45.7	(33)	
		MCD B	46.2	47.7	54.9	64.1	68.4	76.2	78.1	76.0	69.5	61.3	52.4	47.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	11.1	15.3	18.5	18.1	18.4	19.0	18.5	16.8	13.9	9.1	8.1	(35)
MCDBR			10.8	15.8	22.0	25.5	24.8	24.8	26.2	25.6	22.7	17.8	12.0	10.0	(36)	
MCWBR			8.5	10.9	12.0	11.6	10.7	9.6	8.0	9.1	10.2	10.3	8.6	8.0	(37)	
5% WB		MCDBR	10.2	13.9	18.6	22.2	22.0	21.8	21.9	21.5	18.7	15.6	10.7	9.9	(38)	
		MCWBR	8.7	10.1	10.8	10.6	10.5	9.5	7.8	9.2	9.1	9.7	8.5	8.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.331	0.360	0.397	0.415	0.420	0.422	0.427	0.421	0.392	0.384	0.356	0.329	(40)	
taud		2.363	2.296	2.208	2.211	2.231	2.252	2.234	2.277	2.339	2.330	2.346	2.359	(41)		
Ebn at Noon		221	242	252	262	266	266	262	258	253	230	207	203	(42)		
Edh at Noon		21	29	37	41	42	42	42	38	32	28	22	19	(43)		
(44)	All-Sky Solar Radiation	RadAvg	252	485	847	1335	1577	1755	1659	1432	1039	608	285	192	(44)	
(45)		RadStd	38	75	121	170	172	154	185	132	135	64	34	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page